# OM RAJ

**Agricultural Researcher • Environmental Analyst • Software Developer**

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## PROFESSIONAL SUMMARY

Interdisciplinary researcher, environmental analyst, and software developer with an **M.Sc. in Botany (Plant Pathology)** and hands-on project research experience at **IIT Ropar**. Uniquely blends wet-lab experimental rigor (spectrophotometry, chromatography, plant biochemical stress markers, wastewater testing) with full-stack desktop software engineering and digital agriculture knowledge systems. Proven record in executing government-sponsored research projects (PPCB and Punjab Excise Department), authoring peer-reviewed literature, developing production-ready scientific software deployed on the Microsoft Store, and designing digital public platforms. Equipped to drive impact across agri-tech knowledge engineering, environmental compliance monitoring, and computational research pipelines.

## RESEARCH, TECHNICAL & LABORATORY SKILLS

**Analytical & Spectroscopy Instruments:**
ICP-MS (Inductively Coupled Plasma Mass Spectrometry), UV-Visible Spectrophotometer, Gas Chromatography (GC), Ion Chromatography, TVOC Analyzer, Carbon Analyzer, Aerosol Spectrophotometer, Formaldehyde Analyzer.

**Regulatory Auditing & Environmental Compliance:**
Independent third-party monitoring, industrial compliance audits (CPCC, PPCB, Excise Dept.), wastewater testing (CETP, STP, ETP), hazardous waste leachate classification, and technical compliance reporting.

**Software Development & Computational Tech:**
JavaScript, HTML5, CSS3, Electron.js, Node.js basics, REST API integration, Git/GitHub, Visual Studio Code.

**Soil & Solid Waste Characterization:**
Toxicity Characteristic Leaching Procedure (TCLP), acid digestion protocols, solid waste leachate extraction, heavy metal mobility profiling, bio-soil quality assessment, soil organic matter and nutrient testing.

**Agriculture & Plant Pathology:**
Crop disease diagnostics, pathogen identification, IDM, plant stress physiology, antioxidant enzyme assays (SOD, CAT, POD, proline, MDA), biomass/biochar soil amendments, soil fertility profiling.

**Data Science, GIS & Creative Design:**
MS Excel (formulas, linear regression), Power BI dashboards, QGIS, Adobe Photoshop, Canva Pro, Premiere Pro, CapCut, graphical abstracts for scientific journals, SOP authoring.

## PROFESSIONAL RESEARCH & WORK EXPERIENCE

**Project Researcher – Third-Party Environmental Monitoring & Soil Audit Project** Oct 2026 – Present

Dept. of Civil Engineering, IIT Ropar | Client/Sponsor: Chandigarh Pollution Control Committee (CPCC) | PI: Dr. Indramani Dhada

- **Field Sampling & Site Inspection:** Carried out field reconnaissance, verification, and composite soil sampling of legacy-waste bio-remediated material (bio-soil) excavated from Dadumajra Dumping Site and utilized across urban reclamation sites (Sector 47 & Sector 50, Chandigarh).
- **Sample Preparation & Heavy Metal Trace Analysis (ICP-MS):** Prepared solid matrices, carried out acid digestion/mineralization protocols, and performed trace elemental and heavy metal quantification using Inductively Coupled Plasma Mass Spectrometry (ICP-MS).

- **Leachability & Environmental Toxicity (TCLP):** Performed Toxicity Characteristic Leaching Procedure (TCLP) extractions to evaluate heavy metal mobility, leachate contamination hazards, and regulatory compliance per CPCB/hazardous waste guidelines.
- **Soil Quality & Risk Assessment:** Assessed physicochemical parameters (pH, EC, organic content, reactivity), evaluated environmental and public health implications, analyzed data, and contributed to compiling the independent third-party audit report.

### Project Researcher – Sustainable Agriculture & Environmental Monitoring
2025 – Present

Dept. of Civil Engineering, IIT Ropar | Supervisors: Dr. Indramani Dhada, Dr. Neelkanth Nirmalkar | PM: Dr. Harsh Sharma

- **Sustainable Agriculture & Biochar:** Investigated comparative impact of pineapple peel, jackfruit peel, and drek wood biochars on soil physicochemical characteristics, nutrient retention, and crop physiological responses.
- **Plant Biochemical Assays:** Quantified defense enzyme kinetics (SOD, CAT, POD, Proline, MDA) to evaluate plant resilience under abiotic and heavy-metal stress.
- **Wastewater & Industrial Audits (PPCB & Excise Dept.):** Conducted wet-chemical testing across CETPs, STPs, and Sutlej River basin; participated in industrial audits of distilleries evaluating Zero Liquid Discharge (ZLD) performance.

### Postgraduate Dissertation Researcher — River Water Quality & Runoff Assessment
2024 – 2025

Patna University | Supervisor: Prof. Dr. Meenakshi Singh (Head, Dept. of Biotechnology)

- Conducted systematic multi-point field sampling along major ghats of Ganga River at Patna; analyzed physicochemical/biological parameters against WHO, CPCB, and BIS 10500 standards.

### Medical Writing Intern & Freelance Media Specialist
2021 – Present

YuvaIntern (NSDC / Henry Harvin, Nov 2025) & Freelance / MARKAD LIMITED (2021–Present)

- Synthesized biomedical/environmental research reviews with standardized citations (APA, IEEE, Vancouver). Designed graphical abstracts, academic branding, and university election outreach media.

## EDUCATION & ACADEMIC QUALIFICATIONS

| Degree / Program | Specialization / Discipline | Institution / University | Year | Score / CGPA |
|---|---|---|---|---|
| **M.Sc. in Botany** | Plant Pathology | Department of Botany, Patna University | 2025 | **7.56 CGPA** |
| **B.Sc. in Botany** | Plant Sciences & Chemistry | Patna Science College, Patna University | 2023 | 63.50% |
| **Diploma in Computer Applications (DCA)** | Software & Computing Systems | Vinita Computer Education | 2022 | 64.40% |
| **Senior Secondary (Class XII, Science)** | Physics, Chemistry, Biology (PCB) | Shanti Niketan Sr. Sec. School (CBSE) | 2019 | 58.40% |
| **Secondary School (Class X)** | General Science & Mathematics | Sunshine Prep. / High School (CBSE) | 2017 | 81.70% |

## SOFTWARE ENGINEERING & PRODUCTION INNOVATIONS

- **AWE LAB Academic Portal (IIT Ropar):** Architected, designed, and deployed official laboratory web portal for Dr. Indramani Dhada's team at IIT Ropar. (AWE LAB Portal)
- **SNIFF DOI — Academic Literature Discovery Engine (Microsoft Store):** Desktop software indexing 250M+ articles. Features 6-tier research gap synthesis and CSL engine (APA, IEEE, Harvard, Vancouver). (Microsoft Store)
- **WASH (Water Analysis Suite Hub) — Microsoft Store:** Environmental compliance engine automating APHA 24th Ed & BIS 10500 algorithms, spectrophotometer linear regression calibration, and NABL PDF reports. (Microsoft Store)
- **Calculator Luxe & Clock Span (Watch Span):** High-precision scientific desktop calculator & lab reaction kinetics timer published on Microsoft Store.

## PUBLICATIONS & RESEARCH MANUSCRIPTS

- **Peer-Reviewed Journal Article:** Raj, O., & Ambreen, D. (2026). Assessment of Physicochemical Parameters of Water Samples from the Ganga River at Patna. *International Journal of Versatile Research and Analysis (IJVRA)*, 3(4), 784–810. ISSN: 2984-8903. Paper ID: IJVRA2603368.
- **Research Preprint (Air Quality):** Raj, O., & Dhada, I. Spatiotemporal Assessment of Ambient Air Quality in Selected Urban, Tourist and Industrial Regions of Himachal Pradesh, India: A Case Study. *Research Square*. DOI: 10.21203/rs.3.rs-10492149/v1
- **Manuscript in Preparation (Sustainable Agriculture):** Raj, O., & Dhada, I. Comparative Assessment of Pineapple Peel, Jackfruit Peel, and Drek Wood Biochar on Plant Growth, Soil Fertility, and Biochemical Responses. (Evaluates SOD, CAT, POD, proline, MDA kinetics).

## WORKSHOPS & PROFESSIONAL CERTIFICATIONS

- **Fundamentals of Remote Sensing:** NASA "Train - Empower - Advance" Initiative (Sept 2026) – Satellite observations & spatial datasets.
- **Advances in Genome Biology & Gene Regulation:** School of Life Sciences, JNU, New Delhi (June 2025).
- **Marine Data 4 Asia 2025:** Copernicus Marine Service & Mercator Ocean International, EU Program (June 2025).
- **Power BI & AI Interactive Dashboards:** OfficeMaster (Sept 2025).
- **Cancer Genomics, Clinical Research & AI:** Biotecnika (Sept 2025).

## LEADERSHIP & EXTENSION ACTIVITIES

- **Event & Sports Coordinator:** Patna Science College (2023, 2024, 2025). Managed logistics, budgeting, and operations for 3 consecutive Annual Sports Meets. Recipient of *The Catalyst Award (2025)*.
- **Volunteer Coordinator:** Patna University Alumni Meet (2023, 2025). Managed guest reception, VIP protocol, and stage management.

## PROFESSIONAL REFERENCES

**Dr. Indramani Dhada**
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